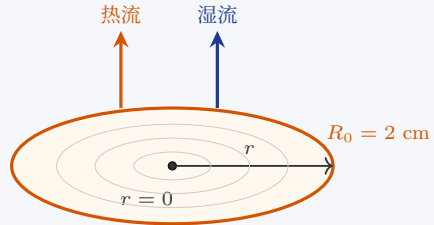


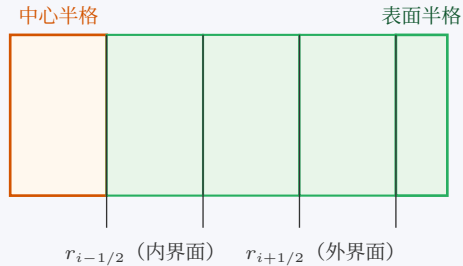
(a) 圆柱截面与边界条件

$$-k\partial_r T = h(T_R - T_\infty), \quad -D\partial_r C = k_m(C_R - C_\infty)$$



$$r = 0: \partial_r T = \partial_r C = 0$$

(b) 径向控制体 ($N = 80, \Delta r = 0.25 \text{ mm}$)



输出点 = 计算点 ($0.1 \text{ cm} = 4\Delta r$)

(c) 时空离散与推进

输出步: $\Delta t_o = 1 \text{ s}$

热步子步: $\Delta t_T = 1/32 \text{ s}$

湿步子步: $\Delta t_C = 0.1 \text{ s}$

